# The Forge — balance report (what the sandbox found)

One Forge system (Territory + Bridges + classes + siege), played in silico. Each number is from the headless Monte Carlo (600 matches, all strategy matchups, both sides, seeded). Two fields: a **symmetric test board** and the **real Lake San Antonio DEM** (the LIB venue). The four heuristics (expander / aggressor / bridger / turtle) plus a **self-play-trained policy (`rl`)** are in the pool.

## Headline findings

**1. Self-play finds an exploit the heuristics never did.** On the symmetric board the trained policy wins **~85%** against the whole heuristic pool — it taught itself to weight the Phase-4 siege tower and forward pressure and to *stop* sitting on owned ground (see the learned weights in `rl/curve.html`). A scripted bot pool would have called this ruleset "balanced"; the RL playtester shows it isn't. *This is the point of RL here — not flashy AI, a better playtester.*

**2. Terrain changes balance. A lot.** The exact same ruleset:

| | symmetric board | real Lake San Antonio |
|---|---|---|
| faction win% (map fairness) | 49 / 51 ✓ | **45 / 55** ⚠ side bias |
| aggressor win% | 56 | **~76** ⚠ dominant |
| rl win% | 85 | 74 |
| turtle win% | 9 | 11 |
| siege fires | 75% | 62% |

The reservoir cuts the real map into bridge-gated halves, and those chokepoints make the **aggressive rush dominant (~76%)** where it was only mild (56%) on open ground. **You cannot balance The Forge once and ship it to every venue — each field needs its own pass.** That's the recurring service the tool provides.

**3. The RL policy doesn't fully transfer between maps (and that's the lesson).** Trained on the symmetric board, `rl` slips from ~85% → ~74% on real terrain — a real transfer gap. A policy (or a balance) tuned on one layout isn't fully valid on the next; you retrain / re-tune per venue. Cheap to find in sim; expensive on a field.

**4. Pure turtling is non-viable (~10%) on both maps** — it can't cross the ravine without bridging. A real design decision, not a bug: give defense a path, or accept that The Forge punishes passivity.

## Lever notes (what to turn, and what it costs)

- **Snowball ~85%** (lead persists). Raising `comebackAid` claws it toward the

trailing side (0 → 0.8 moves comeback% ~4 → ~16) at the cost of softening the reward for early play.

- **Aggression dominant on real terrain** → nerf candidates: `capRate` ↓ (slower flips, defense has time) or `decay` ↑ (overreach slips faster).

- **Siege**: reachable now (fires 60–75%) because of battering + the delay tower.

Wins stay low (~8%) — competent defenders still mostly hold, so score usually decides. That's a healthy "siege is a threat, not a coin flip."

## How to reproduce

``` 
``bash node test/smoke.mjs # invariants (map fairness, lever directions) node rl/selfplay.mjs # retrain the policy → data/rl-policy.json (+ curve) python3 tools/fetch_dem.py # re-pull the venue DEM node tools/build_field.mjs # rebuild the field graph → data/field.json ``
```

Open the live

demo → **Playtest** tab → *Run* to reproduce these on either field.